STATEMENT OF WORK (SOW)
TRAINING AND LOGISTIC SOLUTIONS (TLS)
INTERN PROJECT 2025

Project TEMPER: Temperature Enclosure for Multi-Purpose Environmental Recording

Abstract

Internship Project for 2025 focused on trade study analysis, software interface development, sensor Creo design and final product prepared.

Prepared by: Daejah Walker and Sara Gonzalez Henao

Revision Level	Revision Date	Summary of Change	Pages Affected
D-1.0	02 April 2025	Initial draft of document	All
D-2.0	21 May 2025	Schedule Update	All

1. Project Overview

1.1. Background

Lockheed Martin TLS is a leading supplier of training solutions to the defense industry, with a strong track record of delivering high-quality, innovative training programs that enhance the performance of our military personnel and contribute to organizational success. As part of our ongoing commitment to excellence, we are seeking the expertise of external consultants to assess and forecast the future trajectory of our training initiatives over the next decade. We recognize that maintaining our competitive edge in a rapidly evolving industry requires a forward-thinking approach, as well as the ability to anticipate and prepare for emerging opportunities and threats. By leveraging the insights and expertise of external consultants, we aim to stay ahead of the competition and position ourselves for continued success in the land, air, cyber, and sea military domains.

1.2. Problem Statement

As Lockheed Martin leads a dedicated Line of Business (LOB) for training and sustainment, TLS seeks to continually improve the training system hardware and software to adapt to the ever-changing environment. The Internal Research and Development (IRAD) Container Team, Business Development (BD) identifies the need to design a multi-use enclosure that will provide accurate temperature data. The sensor will need to interact with various LM software applications to display information. Additionally, the system should log temperature data while conforming to designated customer specifications.

2. Project Management

2.1. Scope

The Container Team scope encompasses the design of a multi-use enclosure that provides accurate temperature data, interacts with multiple LM software applications, logs temperature data, and conforms to designated customer specifications.

To achieve this, Project TEMPER scope includes a requirement review to analyze the required specs and MIL standards, a trade study to down-select temperature sensors, software development/demonstration user interface to ensure a seamless interface communication could be ported to the designated software application, and hardware design to propose placement for optimal use within a sample IRAD systems. The scope does not include integration with other systems or hardware currently.

2.2. Objective

The objective is to solution a temperature monitoring system for a multi-use enclosure that provides accurate temperature data and interacts with multiple Lockheed Martin test executive software applications. Key performance indicators (KPIs) will be used to evaluate the project's success, including data accuracy, MOSA (Modular Open Systems Approach) software interface, SWAP-C (Size, Weight, Power and Cost).

2.3. Schedule

TEMPER Schedule

Table 1: Project Milestones

Date	Event
June 10th, 2025	Project Kickoff and SOW Release
June 19th, 2025	Trade study & Sensor Selection Complete
July 21st, 2025	Rough Draft Presentations with Working Product
July 28th, 2025 (2-5 PM EST)	Demonstration Day

Table 2: SME Series

Date	Event
June 12th, 2025	SOW Q/A with LOB Leads (0.5 hours)
June 16th, 2025	SME Series I: Trade Study Analysis
June 23rd, 2025	SME Series II: Software Development
July 8th, 2025	SME Series III: CAD/Creo Design
July 23rd, 2025	SME Series IV: Final Q/A (0.5 hours)

Table 3: Check-Ins

Date	Event
June 19th, 2025	Check In I: Trade Study Analysis Review
July 8th, 2025	Check In II: Material Pickup (After SME Series III)

All events will be offered in person and on teams.

2.4. Requirements

2.4.1. Capability Requirements

2.4.1.1 TAA Compliance

- Solution shall be TAA Compliant.
- TAA Compliance shall be verified using: https://gsa.federalschedules.com/resources/taa-designated-countries/

2.4.1.2 Environmental Standards

- Sensor shall be capable of operating in a MIL-PRF-28800G CLASS I Environment and shall detect temperatures beyond limits within MIL-PRF-28800G (can be found in section 3.8.2, or 3.9.2.1.1, 3.9.1.1.1) in provided specification.

Adobe Acrobat
PDFXML Document

- Software communication shall be via an industry standard communication interface based on the instrument provider API.
- Hardware interface shall be that of a standard PC (may include USB, USB-C, Ethernet, Serial).

2.4.2. Project Requirements

- Each team shall designate one Program Manager who shall have the responsibility of tracking the schedule and ensuring milestones are met and delivered to project coordinators.

- Each team shall designate one Engineering Project Manager who shall have the responsibility of coordinating with Stakeholders and project coordinators.
- Each team shall make available presentation material 24 hours prior to milestone deliveries required from the team.
- Prior to the end of the intern project, each member of the team shall present and/or serve as a liaison (formally or informally with customer) at least once.
- Project demonstration shall take place July 28th, 2025, between 2-5 PM EST and all members of the team are required to participate.

2.5. Assumptions and Constraints

- Stakeholders will be available to provide input for the duration of the project.
- Stakeholders will meet with teams on a periodic basis to check progress.
- Teams will engage with stakeholders early and throughout the project to ensure that their needs are understood and addressed.
- Suggested cost should be within a $0 - $150 range. If desired sensor is outside of limits specified, consult with Stakeholders.
- This competitive project will need to be completed by 7/28/2025 at 9 AM EST for submission.

2.6. Deliverables

- Trade Study Analysis for Down Select of thermocouple
- Requirements Verification Plan
- API for Integration with STORM Lite
- Web-Page GUI
- Functional Product with Sensor
- HW Model of sensor using Creo, or other related modeling software
- Final PowerPoint Presentation

2.7. Acronym List and Definitions

1. LM – Lockheed Martin
2. TLS – Training and Logistic Solutions
3. Stakeholders – Project Leads and Subject Matter Experts

2.8. Trade Study Analysis

Criteria	Sensor A		Sensor B		Sensor C	
	Score (1-5)	Notation	Score (1-5)	Notation	Score (1-5)	Notation
Size						
Weight						
Power						
Cost						
Reliability						
Supportability						
Procurement Lead Time						
TAA Compliant						
Score						

- Meets high-level requirements
- SWaP-C
    - Size
    - Weight and Power
    - Cost
- Reliability (Mean Time Between Failure)
- Long-term supportability
- Procurement lead time

3. Demonstration Day

This section applies to demonstration day only including requirements, event schedule and judging criteria.

3.1. Related Requirements

- Each team shall make available presentation material 24 hours prior to milestone deliveries required from the team.
- Project demonstration shall take place July 28th, 2025, between 2-5 PM EST and all members of the team are required to participate.

3.2. Schedule

Activity	Time	Responsible Individuals
Opening/Introductions	2 PM - 2:15 PM	Sara and Daejah
Group 1 Presentation with Q/A	2:20 PM – 3 PM	Group 1 members
Group 2 Presentation with Q/A	3 PM – 3:40 PM	Group 2 members
Group 3 Presentations with Q/A	3:40 PM – 4:20 PM	Group 3 members
Judging/Break	4:20 PM – 4:45 PM	Judges
Awards and Closing	4:45 PM – 5 PM	Sara and Daejah

3.3. Criteria

Technical – 100 points

- Trade Study Analysis and Selection Process
- HW Design in Creo
- Communication with Sensor
- Web-page GUI
- Accuracy of results
- Fully Integrated System

Programmatic – 100 points

- Requirement completion and verification
- Presentation neatness and creativity
- Presentation engagement

Incentives – Additional 20 points

- Thermal Analysis of Sensor